

Software Engineering

Chapter 6 & 7: Requirements Modeling

A process is a roadmap for building high-quality software.

Contents

- What is Requirements?
- Requirements Engineering (RE) Process
- Steps of RE
- Software Requirements Specification (SRS)

Requirements Modeling

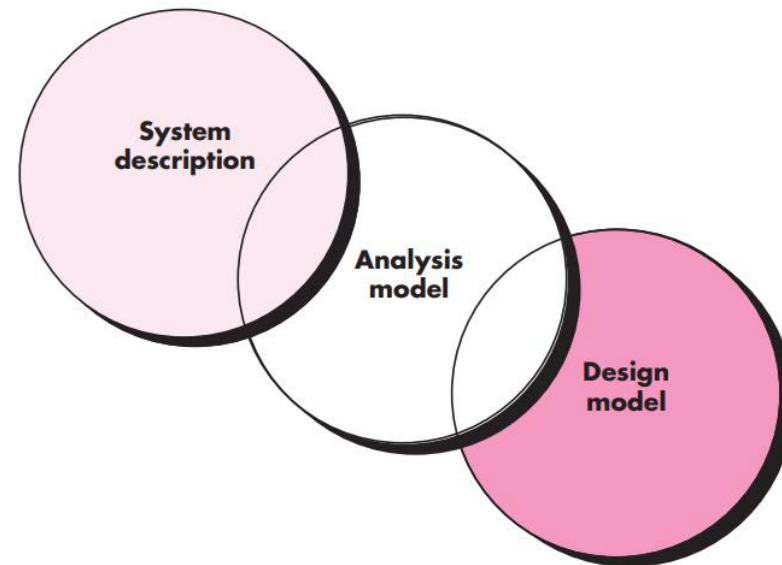
- ***Requirements Modelling (RM)*** is the process of **representing and visualizing** what the system should do, before you actually start designing or coding it.
- It is a ***set of models*** that is the ***first technical representation*** of a system.
- When we model requirements, we try to **understand the system from different perspectives:**
 - What users need?
 - How data will flow?
 - What processes will occur?
 - How the system will behave under different conditions?

Requirements Analysis

- The requirements modeling action results in one or more of the following **types of models**:
 - ***Scenario based models***: Represents requirements from the point of view of various system actors.
 - ***Data Models***: It represents the information domain for the problem.
 - ***Class oriented models***: It represents object-oriented classes (attributes and operations) and the manner in which classes collaborate to achieve system requirements.
 - ***Flow-oriented models***: It represents the functional elements of the system and how they transform data as it moves through the system.
 - ***Behavioral models***: It depicts how the software behaves as a consequence of external “events”

Requirement Analysis (Cont..)

- These models provide a software designer with information that can be ***translated to architectural, interface, and component-level designs.***
- The ***requirements model*** as a bridge between the ***system description*** and the ***design model***
- .

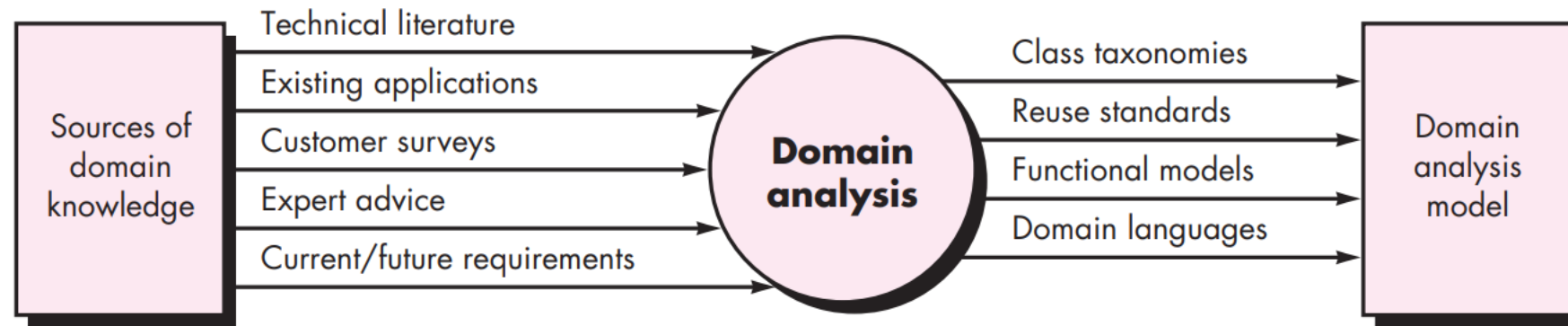


Objectives of Requirements Modeling

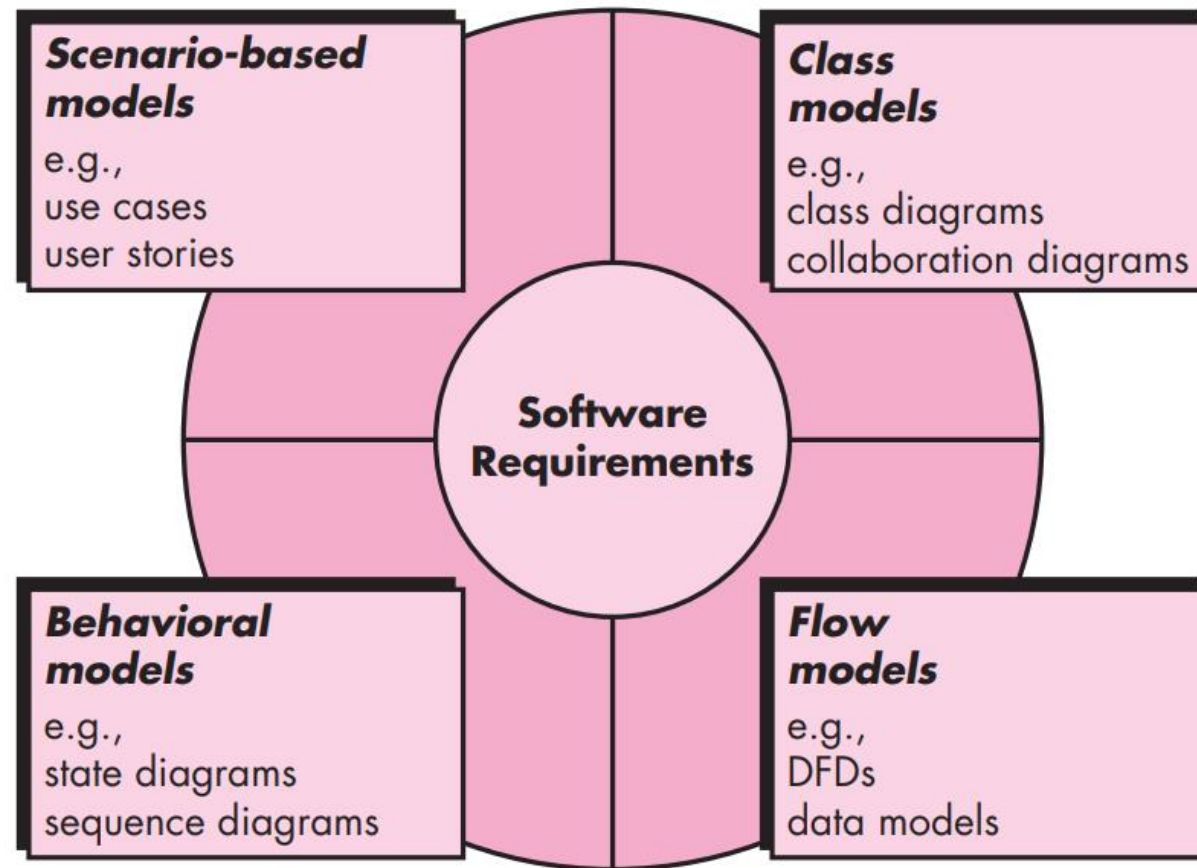
- Throughout requirements modeling, your primary focus is on what, not how.
- The requirements model must achieve **three** primary objectives:
 - (1) to describe what the **customer requires**,
 - (2) to establish a **basis** for the creation of a **software design**, and
 - (3) to define a set of requirements that can be **validated** once the software is built.

Domain Analysis

FIGURE 6.2 Input and output for domain analysis



Requirements Modeling Approaches



1. Scenario Based Modeling

- Creating a Preliminary Use Case
- Refining a Preliminary Use Case
- ***Writing a formal Use Case***
- ***Use Case Diagram***
- UML Models for Representing Use Case (Supplement of the Use Case)
 - ***Activity Diagram***
 - ***Swimlane Diagram***

Use Case Template

SAFEHOME



Use Case Template for Surveillance

Use case: Access camera surveillance via the Internet—display camera views (ACS-DCV)

Iteration: 2, last modification: January 14 by V. Raman.

Primary actor: Homeowner.

Goal in context: To view output of camera placed throughout the house from any remote location via the Internet.

Preconditions: System must be fully configured; appropriate user ID and passwords must be obtained.

Trigger: The homeowner decides to take a look inside the house while away.

Scenario:

1. The homeowner logs onto the *SafeHome Products* website.
2. The homeowner enters his or her user ID.
3. The homeowner enters two passwords (each at least eight characters in length).
4. The system displays all major function buttons.
5. The homeowner selects the “surveillance” from the major function buttons.
6. The homeowner selects “pick a camera.”
7. The system displays the floor plan of the house.
8. The homeowner selects a camera icon from the floor plan.
9. The homeowner selects the “view” button.
10. The system displays a viewing window that is identified by the camera ID.
11. The system displays video output within the viewing window at one frame per second.

Exceptions:

1. ID or passwords are incorrect or not recognized—see use case **Validate ID and passwords**.
2. Surveillance function not configured for this system—system displays appropriate error message; see use case **Configure surveillance function**.
3. Homeowner selects “View thumbnail snapshots for all camera”—see use case **View thumbnail snapshots for all cameras**.
4. A floor plan is not available or has not been configured—display appropriate error message and see use case **Configure floor plan**.
5. An alarm condition is encountered—see use case **Alarm condition encountered**.

Priority: Moderate priority, to be implemented after basic functions.

When available: Third increment.

Frequency of use: Moderate frequency.

Channel to actor: Via PC-based browser and Internet connection.

Secondary actors: System administrator, cameras.

Channels to secondary actors:

1. System administrator: PC-based system.
2. Cameras: wireless connectivity.

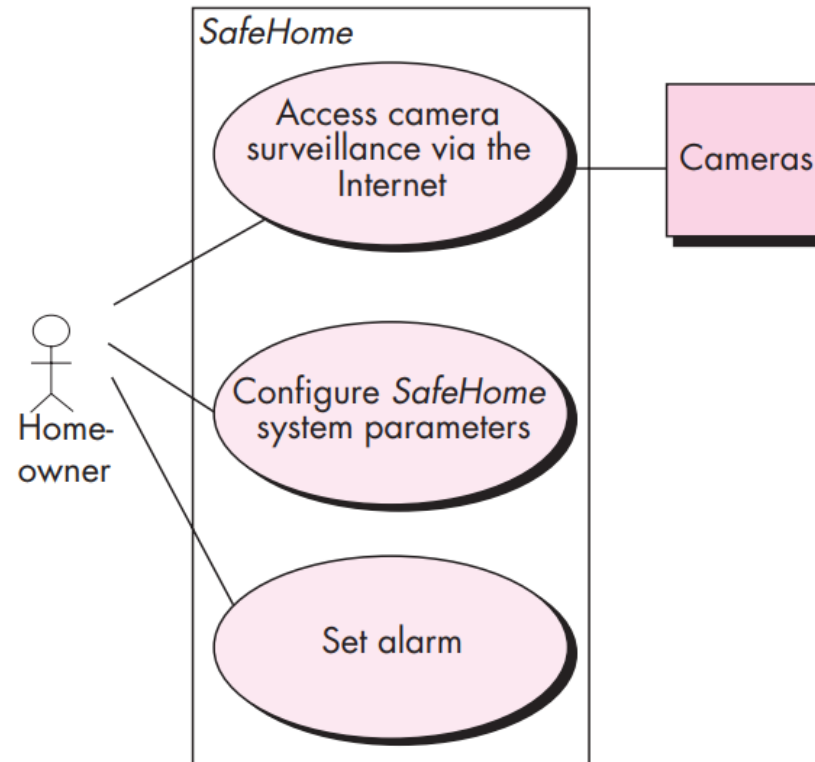
Open issues:

1. What mechanisms protect unauthorized use of this capability by employees of *SafeHome Products*?
2. Is security sufficient? Hacking into this feature would represent a major invasion of privacy.
3. Will system response via the Internet be acceptable given the bandwidth required for camera views?
4. Will we develop a capability to provide video at a higher frames-per-second rate when high-bandwidth connections are available?

Use case Diagram

FIGURE 6.4

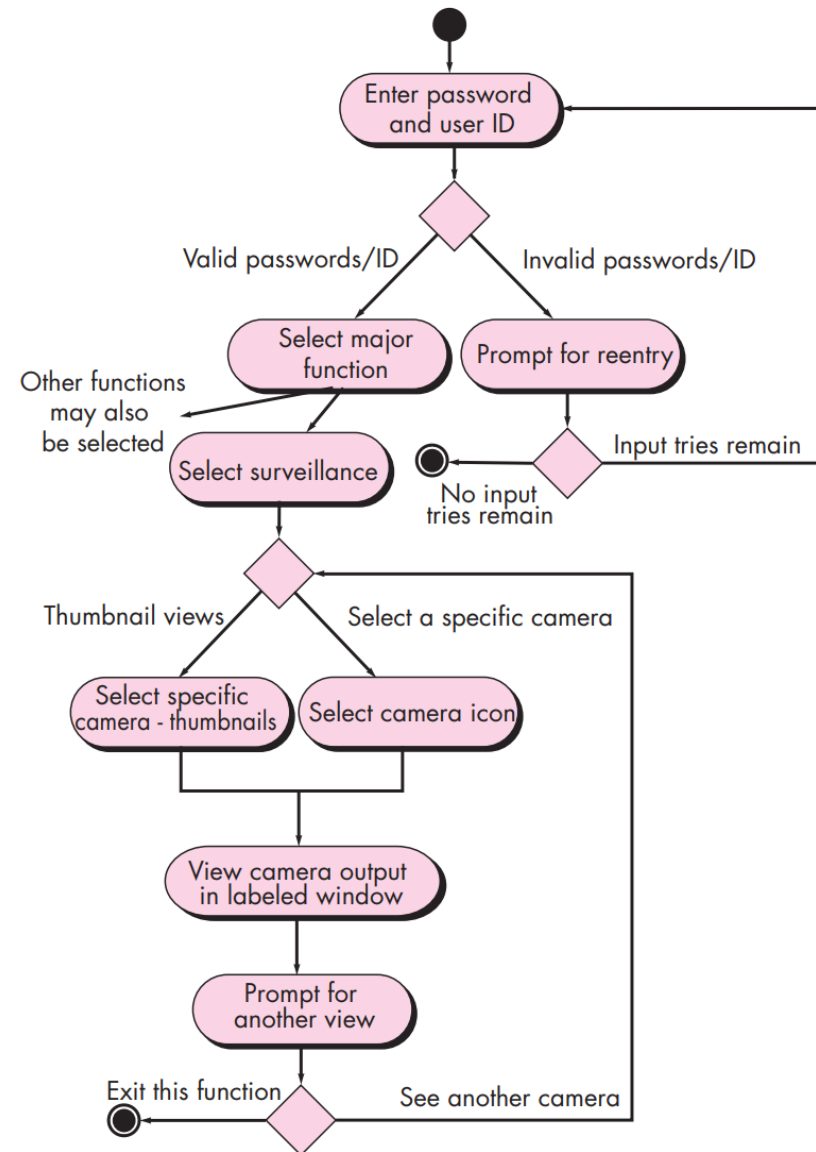
Preliminary
use-case
diagram for
the *SafeHome*
system



Activity Diagram

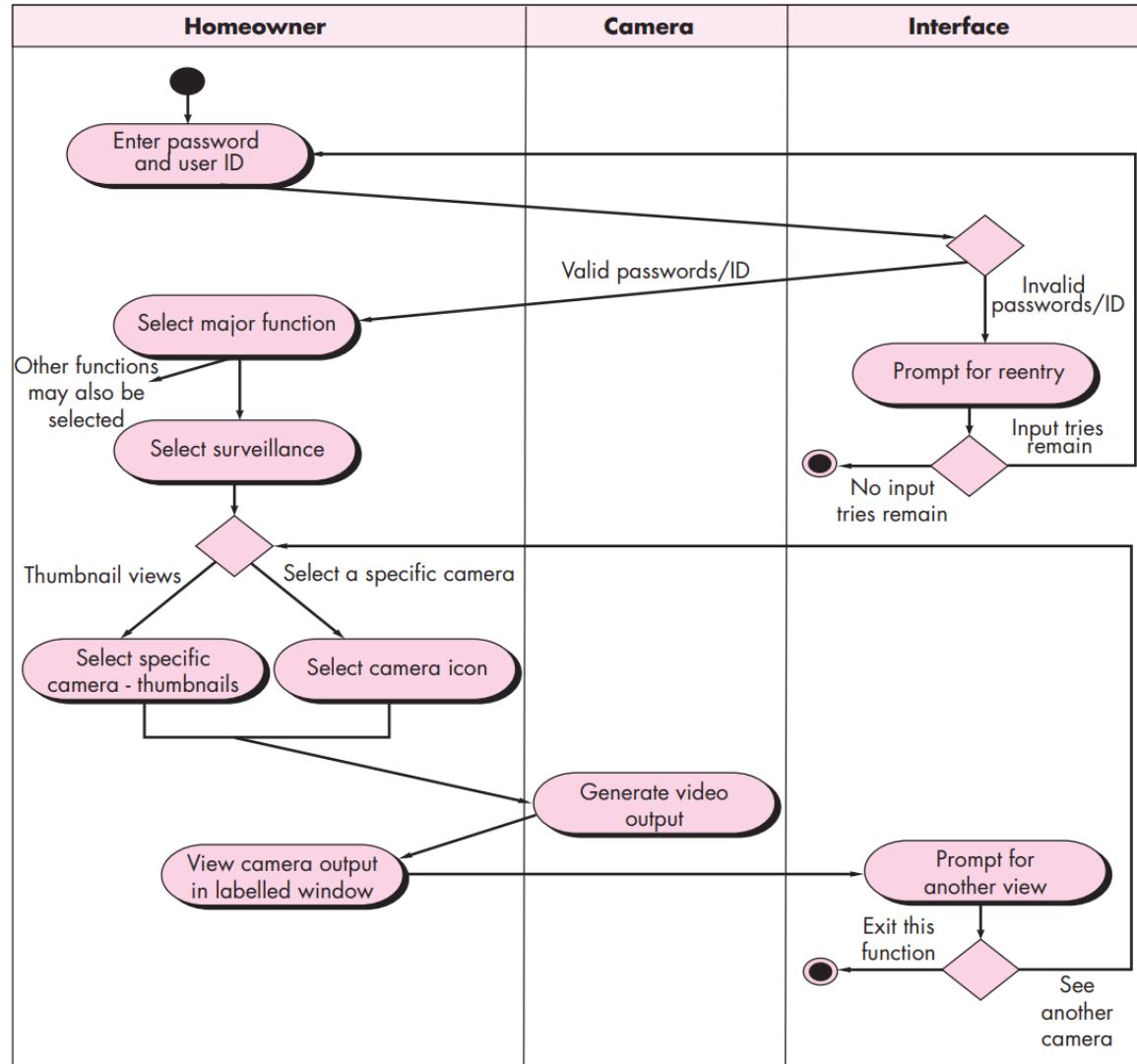
FIGURE 6.5

Activity diagram for Access camera surveillance via the Internet—display camera views function.



Swimlane Diagram

FIGURE 6.6 Swimlane diagram for Access camera surveillance via the Internet—display camera views function



2. Data Modeling

- ERD
 - Data Objects (Entity)
 - Data Attributes
 - Relationships
- Schema

3. Class Oriented Modeling

- Class-based modeling represents:
 - **objects** that the system will manipulate
 - **operations** (also called methods or services) that will be applied to the objects to effect the manipulation
 - **relationships** (some hierarchical) between the objects
 - **collaborations** that occur between the classes that are defined.
- The elements of a class-based model include **classes and objects**, **attributes**, **operations**, **CRC models**, *collaboration diagrams and packages*.

Identifying Analysis Classes

- **External entities** (e.g., other systems, devices, people) that produce or consume information
- **Things** (e.g., reports, displays, letters, signals) that are part of the information domain for the problem
- **Occurrences or events** (e.g., a property transfer or the completion of a series of robot movements) that occur within the context of system operation
- **Roles** (e.g., manager, engineer, salesperson) played by people who interact with the system
- **Organizational units** (e.g., division, group, team) that are relevant to an application
- **Places** (e.g., manufacturing floor or loading dock) that establish the context of the problem and the overall function
- **Structures** (e.g., sensors, four-wheeled vehicles, or computers) that define a class of objects or related classes of objects

Samples classes on Library Management System

Category	Description	Example (Library System)
External Entities	People or systems that interact with the system	Librarian, Member, Payment Gateway
Things	Information objects	Book Record, Borrowing Slip
Occurrences/Events	Activities or system triggers	Book Issued, Book Returned
Roles	Responsibilities or user types	Member, Librarian, Administrator
Organizational Units	Divisions or departments	Library Department, Accounts Section
Places	Physical locations	Issue Desk, Library Hall
Structures	Physical or logical objects	Barcode Scanner, Database Server

Performing Grammatical Parse (Identify classes)

The SafeHome security function *enables* the homeowner to *configure* the security system when it is *installed*, *monitors* all sensors *connected* to the security system, and *interacts* with the homeowner through the Internet, a PC, or a control panel.

During installation, the SafeHome PC is used to *program* and *configure* the system. Each sensor is assigned a number and type, a master password is programmed for *arming* and *disarming* the system, and telephone number(s) are *input* for *dialing* when a sensor event occurs.

When a sensor event is *recognized*, the software *invokes* an audible alarm attached to the system. After a delay time that is *specified* by the homeowner during system configuration activities, the software dials a telephone number of a monitoring service, *provides* information about the location, *reporting* the nature of the event that has been detected. The telephone number will be *redialed* every 20 seconds until telephone connection is *obtained*.

The homeowner *receives* security information via a control panel, the PC, or a browser, collectively called an interface. The interface *displays* prompting messages and system status information on the control panel, the PC, or the browser window. Homeowner interaction takes the following form . . .

Potential Classes

Potential Class

homeowner

sensor

control panel

installation

system (alias security system)

number, type

master password

telephone number

sensor event

audible alarm

monitoring service

General Classification

role or external entity

external entity

external entity

occurrence

thing

not objects, attributes of sensor

thing

thing

occurrence

external entity

organizational unit or external entity

Class Selection Criteria (Coad & Yourdon)

- **Retained information:** The potential class will be useful during analysis only if information about it must be remembered so that the system can function.
- **Needed services:** The potential class must have a set of identifiable operations that can change the value of its attributes in some way.
- **Multiple attributes:** During requirement analysis, the focus should be on "major" information; a class with a single attribute may, in fact, be useful during design, but is probably better represented as an attribute of another class during the analysis activity.
- **Common attributes:** A set of attributes can be defined for the potential class and these attributes apply to all instances of the class.
- **Common operations:** A set of operations can be defined for the potential class and these operations apply to all instances of the class
- **Essential requirements :** External entities that appear in the problem space and produce or consume information essential to the operation of any solution for the system will almost always be defined as classes in the requirements model.

Potential Classes (SafeHome)

Potential Class

homeowner

sensor

control panel

installation

system (alias security function)

number, type

master password

telephone number

sensor event

audible alarm

monitoring service

Characteristic Number That Applies

rejected: 1, 2 fail even though 6 applies

accepted: all apply

accepted: all apply

rejected

accepted: all apply

rejected: 3 fails, attributes of sensor

rejected: 3 fails

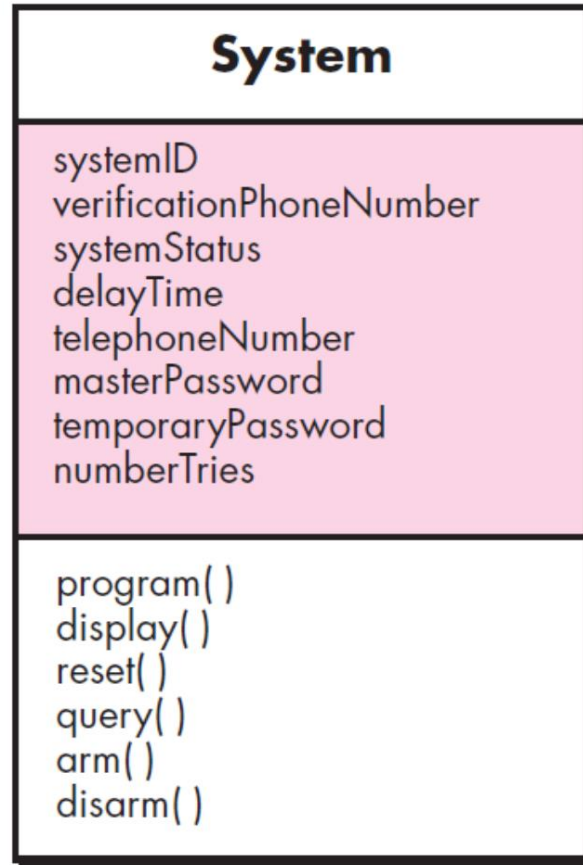
rejected: 3 fails

accepted: all apply

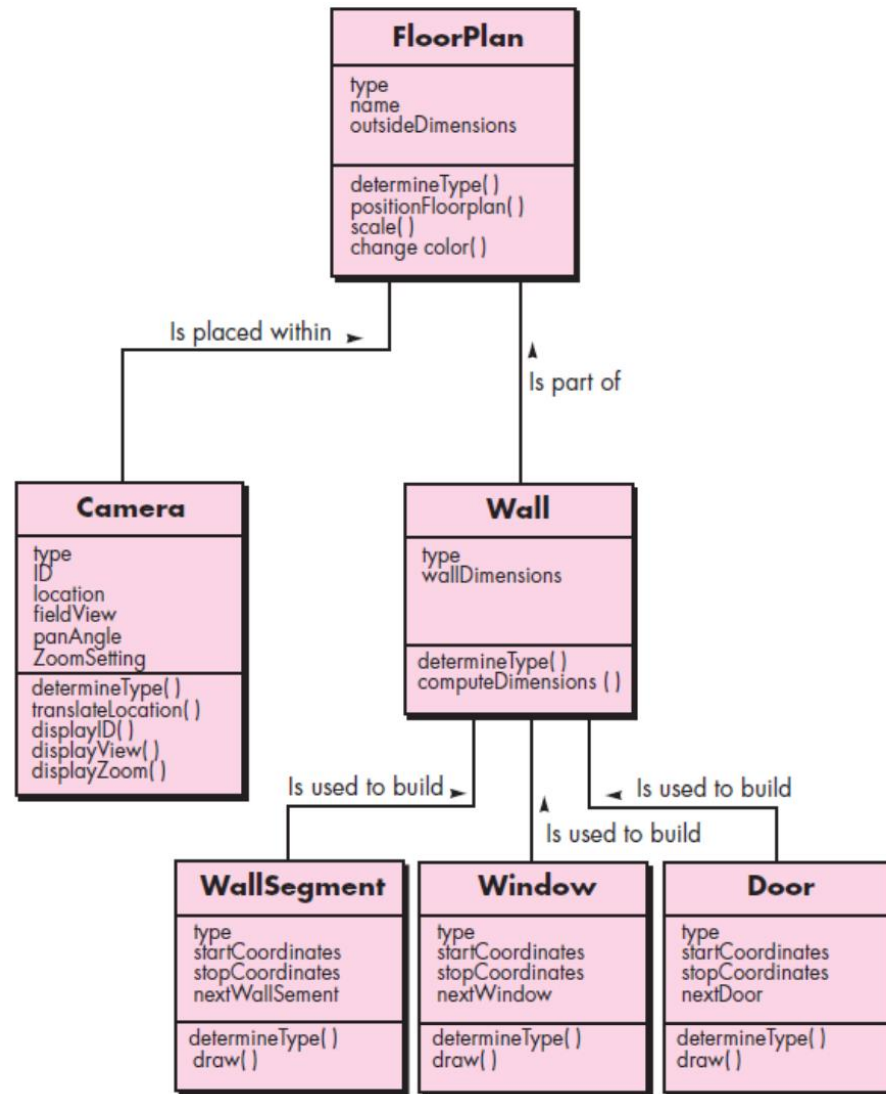
accepted: 2, 3, 4, 5, 6 apply

rejected: 1, 2 fail even though 6 applies

System Class for SafeHome (Class Card)



Class Diagram for FloorPlan



4. Data Flow Modeling

- Data Flow Diagram

5. Behavioral Modeling

- State Transition Diagram
- Sequence Diagram

References

- Chapter 6: Requirements Modeling(Pressman - 7 edition)

Thank you